

The effects of experimentally induced emotions on revising common vaccine misconceptions

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Abstract

Knowledge revision is the process of updating incorrect prior knowledge in light of new, correct information. Although theoretical and empirical knowledge has advanced regarding the cognitive processes involved in revision, less is known about the role of emotions, which have shown inconsistent relations with key revision processes. This study examined the effects of experimentally induced emotions on online and offline knowledge revision of vaccination misconceptions. Before reading refutation and non-refutation texts, 96 individuals received a positive, negative, or no emotion induction. Findings showed that negative emotions, more than positive emotions, resulted in enhanced knowledge revision as indicated by greater ease of integrating correct information during reading and higher comprehension test scores after reading. Findings are discussed with respect to contemporary frameworks of knowledge revision and emotion in reading comprehension and implications for educational practice.

Keywords

Emotions; emotion induction; knowledge revision; refutation texts; misconceptions; vaccination

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On 13 January 2020, the World Health Organization (WHO, 2020) released a list of what it saw as the urgent challenges facing global health for the coming decade. Among the list—which included devastating scourges like HIV, climate change, and, presciently, a pandemic—is trust and misinformation around vaccines. As the WHO (2019) reported, “Vaccine hesitancy—the reluctance or refusal to vaccinate despite the availability of vaccines—threatens to reverse progress made in tackling vaccine-preventable diseases.” Unique among these challenges, vaccine hesitancy is distinguished by the fact that it is a fundamentally psychological phenomenon, formed by people’s beliefs, emotions, and the (mis)information to which they are exposed.

Unfortunately, recent findings show that some attempts to correct vaccine misinformation are ineffective or even backfire, strengthening misconceptions (Nyhan, Reifler, Richey, & Freed, 2014; Pluviano, Watt, & Della Sala, 2017; Schmidt, Zollo, Scala, Betsch, & Quattrocioni, 2018; Trevors & Duffy, 2020). There is an urgent need to continue research on understanding effective ways to combat vaccine misinformation because the link between individuals’ beliefs and global health is remarkably direct. For instance, roughly a third of US states allow personal belief

exemptions from school immunisation requirements (National Conference of State Legislatures, 2017), the use of which have been linked to outbreaks of vaccine-preventable diseases (Immunization Action Coalition, 2016; Omer et al., 2006; Phadke, Bednarczyk, Salmon, & Omer, 2016). Furthermore, developing effective methods to overcome vaccine misconceptions is likely necessary to attain an immunisation rate between 55% and 82% of the population estimated to be needed to gain control over the COVID-19 pandemic (DeRoo, Pudalov, & Fu, 2020). Thus, the persistence of misinformation in decision-making about immunisation even among a minority of the population may directly lead to serious and irreversible harm to patients and other vulnerable communities.

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One reason for the exceptional persistence of vaccine misinformation in memory may be its prominent emotional dimension (Vosoughi, Roy, & Aral, 2018; Zollo et al., 2015). For example, vaccine misinformation may involve patients' or new parents' confusion over conflicting information, fear of serious adverse side effects, or anger over infringement of freedoms in following a mandated vaccination schedule (Dixon & Clarke, 2013; Hornsey, Harris, & Fielding, 2018; Sotiriadis et al., 2012). Despite the obvious presence and potential impact of emotions, no research to our knowledge has experimentally examined the causal effects of emotions during attempted corrections of vaccine-related misconceptions.

This study addressed this issue by experimentally inducing various affective states to examine the impact of emotion as individuals revise their incorrect knowledge about vaccines and learn correct knowledge. We draw on a recent theoretical framework of knowledge revision and the cognitive processes it entails, as well as a recent theoretical framework on emotion that may inform its potential influence on knowledge revision.

Correcting misconceptions with refutations

Over several decades, researchers have sought to develop interventions to more effectively combat misconceptions. As a result of this research, consistent empirical support has been amassed for refutation texts as one form of successful intervention. Refutation texts are typically short documents that target a misconception for revision by identifying a single erroneous belief, refuting it, and presenting the best available evidence as an alternative explanation (Hynd, 2001). In a majority of cases, texts that possess these elements result in greater knowledge revision in comparison to texts that do not (Ariasi, Hyönä, Kaakinen, & Mason, 2017; Diakidoy, Mouskounti, Fella, & Ioannides, 2016; Jones, Johnson, & Campbell, 2015; Kendeou, Smith, & Brien, 2013; Kendeou, Walsh, Smith, & Brien, 2014; Kendeou & van Den Broek, 2007; Thacker et al., 2020; Trevors, Kendeou, Bråten, & Braasch, 2017; Trevors, Kendeou, & Butterfuss, 2017; van den Broek, 2010; van Loon, Dunlosky, van Gog, van Merriënboer, & de Bruin, 2015).

Refutation texts attain their success via engendering the cognitive processes necessary to revise incorrect prior knowledge. Kendeou and O'Brien (2014) proposed the Knowledge Revision Components (KReC) framework to describe the conditions and mechanisms by which refutations promote these cognitive processes leading to revision. First, once knowledge is encoded in memory, correct or incorrect, it is assumed to remain indefinitely (encoding principle). Second, consistent with general models of cognitive architecture, the influence of knowledge is assumed to vary as a function of its activation, which spreads in a passive and unrestrictive manner (passive activation principle).

Taken together, these two principles predict that in the absence of educational interventions, misconceptions will maintain their potential to be reactivated and detrimentally influence thinking and decision-making.

The KReC framework posits that refutation texts serve as an effective intervention due to the cognitive mechanisms they engender, including co-activation, integration, and competitive activation. The co-activation principle refers to the prerequisite that a misconception is activated simultaneously with the new, correct information that refutes it. Co-activation enables integration (integration principle), which refers to the requirement that correct information and its connection to a misconception is integrated in memory. The integration of correct and incorrect information in memory creates a conflict between concepts that is resolved by the competing activation principle: the outcome of knowledge revision is determined by whether correct or incorrect information receives more activation in memory. Thus, as individuals read refutation texts, they are presented with content that first activates a prior misconception ("Many people believe that . . .") that is then paired with new information that refutes it ("However, that idea is incorrect"). The amount of activation that is allocated between correct and incorrect information is determined by the quantity and quality of supporting information. If the explanation provided by refutation texts is sufficiently detailed, then the correct knowledge advocated in the refutation draws more activation towards it at the expense of the misconception and thus becomes dominant in the integrated mental representation. As a result, misconceptions will have a lower likelihood of reactivation and influence over future comprehension, learning, or decision-making.

The KReC framework has been empirically supported by research that examined both the outcomes from reading refutation texts as well as the moment-by-moment cognitive processing during reading. Analysis of unfolding cognitive processes using think-aloud showed evidence of co-activation when individuals initially processed the refutation (Kendeou, Butterfuss, Kim, & Van Boekel, 2019; McCrudden & Kendeou, 2014; van den Broek & Kendeou, 2008) and eye-tracking patterns revealed integrative processing (Ariasi et al., 2017). In a series of experiments, Kendeou and colleagues (2013, 2014) manipulated the strength of supporting correct explanations in refutations to demonstrate the influential effect of competing activation on knowledge revision: when the number of explanations increased, the likelihood of successful revision also increased. Other research that has likewise manipulated refutation content to promote more elaborative cognitive engagement has shown greater likelihood of knowledge revision with greater levels of activation allocated towards processing correct content (Danielson, Sinatra, & Kendeou, 2016; Rich, van Loon, Dunlosky, & Zaragoza, 2017). After reading, refutation texts result in greater learning gains (Donovan, Zhan, & Rapp, 2018; Trevors & Muis, 2015).

higher quality explanations produced by readers (Will, Masad, Vlach, & Kendeou, 2019), and learning that transfers to novel, applied situations (Beker, Kim, Van Boekel, van den Broek, & Kendeou, 2019; Prinz, Golke, & Wittwer, 2018). Meta-analytic research has confirmed the consistent benefits of refutation texts for knowledge revision (Guzzetti, Snyder, Glass, & Gamas, 1993; Tippet, 2010).

In sum, refutation texts are an effective tool to correct misconceptions. However, little is known regarding how knowledge revision processes interact with emotions. This is a consequential gap in knowledge given that emotion is involved in the formation, spread, and persistence of misconceptions on particularly significant topics, such as vaccinations (Berger & Milkman, 2012; British Broadcasting Corporation [BBC], 2014; Manca, 2018; Martel, Pennycook, & Rand, 2019; Nugier, Limousi, & Lydié, 2018). It is, therefore, imperative to understand how prevalent tools for correction interact with individuals' emotions. To this end, we draw on the literature on the relations between emotion, refutation texts, and core cognitive processes outlined in the KReC framework.

Inconsistent relations between emotions and knowledge revision

Several researchers have argued that emotional valence—which can be either positive/pleasant or negative/unpleasant—is intuitively aligned with positive/successful or negative/unsuccessful learning outcomes from knowledge revision interventions. For instance, higher levels of positive emotions like enjoyment and curiosity predict knowledge revision indirectly via approach goal orientation, motivation, engagement, self-reported elaboration, and critical thinking (Muis et al., 2018; Taasobshirazi, Heddy, Bailey, & Farley, 2016). Similarly, reading a refutation text is associated with higher levels of positive emotions, lower negative emotions, higher conceptual knowledge, and more favourable attitudes consistent with the presented evidence (Broughton, Sinatra, & Nussbaum, 2013; Heddy, Danielson, Sinatra, & Graham, 2017; Thacker et al., 2020). Conversely, when corrective messages conflict with valued beliefs, negative activating emotions like anger, anxiety, and stress are experienced, which may lead to derogation of corrective messages and lower learning gains (Nauroth, Gollwitzer, Bender, & Rothmund, 2014; Trevors, 2020; Trevors, Muis, Pekrun, Sinatra, & Winne, 2016; Weeks, 2015).

Others have argued for the inverse relationship, where negative emotions promote successful outcomes (e.g., faster integration, improved memory) and positive emotions predict unsuccessful outcomes in knowledge revision. Some empirical findings support this paradoxical relationship between emotional valence and processes integral to knowledge revision. Compared with positive emotions, experimentally induced negative emotions caused individuals to

become more sensitive to and adapt to discrepancies between stimuli in general (Schuch & Koch, 2015) and more sensitive to discrepancies between the unfolding semantic context of a text and the meaning of new, conflicting information in particular (Pinheiro et al., 2013), which reflects co-activation and integration processes. In such cases, it is theorised that discrepant information itself carries a negative valence (Dignath, Eder, Steinhäuser, & Kiesel, 2020), which when aligned with an individuals' negative affective state, enables a processing advantage for emotionally congruent information and may, therefore, implicate competing activation advantage for corrective information. Rather than being more difficult to correct, retractions of negative misinformation are as effective as retractions of non-emotional misinformation (Ecker, Lewandowsky, & Apai, 2011) and more effective when there is an emotional congruency between negative misinformation and general mood of depressive ruminators (Chang, Ecker, & Page, 2019). In a similar vein, negative emotional words embedded in refutation texts augmented cognitive revision processes and attitude change over positive or neutral words (Sangalang, Ophir, & Cappella, 2019; Trevors & Kendeou, 2020), which may reflect how negative emotion specifically resulted in heightened activation for correct concepts. Automated or self-evaluated analysis of facial expressions after presentation of contradictory information showed higher levels of negative affect, including sadness, disgust, and confusion, which in turn resulted in higher levels of learning (D'Mello, Lehman, Pekrun, & Graesser, 2014) and knowledge revision (Chiu, Liaw, Yu, & Chou, 2019). In addition, Scrimin and colleagues (2019) found that individuals' ability to flexibly change conceptions in response to a challenging situation was more likely with a corresponding upregulation of a moderate physiological stress response than without.

To make sense of these seemingly contradictory findings, Bohn-Gettler (2019) argued that we need greater specificity in understanding emotion research. To this end, the process, emotion, and task (PET) framework organised empirical findings on emotion not only according to the type of emotion (e.g., positive, negative) but also the type of task and comprehension process involved. Findings may appear to be inconsistent or conflicting across studies if only valence is attended to and comparisons are made between different comprehension tasks or processes. This reorganisation and focus on specificity generated several relevant hypotheses. For instance, congruency effects may sometimes occur, such that performance on memory tasks is facilitated for information that shares the same positive or negative valence as the emotional state of an individual (Bower, 1981; Schwarz & Skurnik, 2003). In addition, positive emotions facilitate performance on constructive tasks that involve the generation of new connections or ideas. In contrast, negative emotions facilitate reconstructive tasks that involve closely attending to and reproducing external content. Building on research from Fiedler and

Beier (2014), the PET framework also contended that emotional states align with two broader cognitive strategies found in conceptual change literature: accommodation and assimilation (Forgas, 2013; Lord, Ross, & Lepper, 1979; Posner, Strike, Hewson, & Gertzog, 1982). These cognitive processes are focused on the interaction between internal memory representations, like misconceptions, and new external information, as may be introduced by a refutation text. Accommodation is a bottom-up, reproductive process focused on changing internal representations to reflect new, external information. In contrast, assimilation is a top-down, generative process of modifying the perception of external information to fit pre-existing knowledge. Importantly, negative emotion facilitates accommodation via promoting focused attention to details and enhanced, accurate memory, whereas positive emotion facilitates assimilation via promoting broad attention and generative and elaborative processes (Bohn-Gettler, 2019). In the specific context of learning from a refutation text, focused attention on the information it contains, and accurate memory thereof, are integral to successful knowledge revision. Thus, insofar as negative emotion facilitates accommodation, it appears to be a state more conducive to knowledge revision from refutation texts in comparison to the assimilation that may arise from positive emotion (Palmer, 2003).

In sum, there exists inconsistent and paradoxical relations between emotion valence and knowledge revision outcomes that may depend on the type of task and processes involved (e.g., constructive vs reconstructive; assimilative vs accommodative). There are several studies that show retrospectively self-reported positive emotions like curiosity and enjoyment are associated with successful updating of prior knowledge, whereas self-reported negative emotions like anger and anxiety are associated with failure to update. However, there are theoretical and empirical reasons to believe that the inverse relationship exists between emotion valence and knowledge revision outcomes when more objective in situ measures of emotions are used. These studies show that negative emotion facilitates processes fundamental to knowledge revision, including co-activation, integration, and competing activation. Crucially, any factor that influences the competition for activation between correct and incorrect concepts will likewise affect the probability and strength of knowledge revision. Given these inconsistencies in past findings, the objective of the current research is to help clarify the relation between emotion and knowledge revision.

Present research

We sought to determine the effects of in situ emotion on knowledge revision processes and outcomes with refutation texts. To accomplish this, we experimentally induced positive, negative, or a no/neutral emotion in individuals

(between-subjects) prior to reading refutation texts on vaccines. All participants were presented with nine texts across three refutation conditions (within-subjects): texts that refuted a common vaccine misconception and explained the correct concept (Refutation + Explanation); texts that refuted but did not explain (Refutation-Only); and texts that neither refuted nor explained (No Refutation + No Explanation control). These three conditions allowed us to isolate and examine the effects of text structure on specific knowledge revision processes such as co-activation, integration, and competing activation, and most important, to examine the interaction between these processes and emotion.

We estimated knowledge revision in two ways. First, we measured reading times on phrases that presented the correct information and were identical across all text conditions. Reading times provide a reliable index of online comprehension relevant to knowledge revision (Kaakinen, 2017; Kendeou et al., 2013; Murray & Engle, 2005), which have been additionally validated with other concurrent measures (Kendeou et al., 2019). In line with previous research, we assumed that the speed of reading correct-outcome phrases reflects the ease of integrating that information into the evolving mental representation. Thus, slower times indicate continued activation and influence of prior misconceptions, whereas faster times indicate correct knowledge has drawn activation away from misconceptions. The second method we used to measure knowledge revision was via participants' responses and explanations for their responses on a knowledge test administered before and after reading.

Hypotheses related to emotion induction

Given that emotion in this study was experimentally induced and not retrospectively self-reported, we predicted that results would be aligned with past research that shows advantageous effects for negative over positive emotions (Bohn-Gettler, 2019). Specifically, we predicted that participants who received negative emotion induction will have superior knowledge revision processes and outcomes demonstrated via fastest reading times and highest post-test scores overall; positive induction will result in the slowest times and lowest scores; and neutral induction will fall in between.

In addition, we predicted that the advantage for negative emotion induction on revision processes and outcomes may be in part attributable to the effect of affective congruency between negative induction and the negative valence carried by conflicting information (Dignath et al., 2020). To support this hypothesis, we conducted a second study that collected the perceived emotional valence of refutation content relative to analogous non-refutation content to determine whether the former is in fact perceived more negatively. If so, such negatively valenced information

may be more attended to and remembered by individuals experiencing a negative state (Bohn-Gettler, 2019; Bower, 1981; Schwarz & Skurnik, 2003) and may, therefore, confer a competitive activation advantage for corrective information (Kendeou et al., 2014).

Hypotheses related to text structure

Consistent with the KReC framework (Kendeou et al., 2014), we predicted that Refutation + Explanation texts will result in the fastest reading times and highest scores overall; No Refutation + No Explanation texts will result in the slowest times and lowest scores; and Refutation-Only will fall in between.

Hypothesis related to emotion $\times$ text interaction

Finally, consistent with the PET framework (Bohn-Gettler, 2019), we also expected that the effects of emotion will be stronger in the Refutation-Only condition. Refutations without explanations facilitate co-activation of correct and incorrect information but do not privilege or constrain activation on the correct concept. As such, responses to Refutation-Only texts are likely to be more heterogeneous than either of the other two text conditions and depended on other influential factors. With fewer constraints provided externally by these texts, emotions are theorised to have a stronger effect in directing activation and thus on the processes and outcomes of revision.

Method

Participants

Participants were 96¹ undergraduate students at a large public university in the Midwest. Their average age was 19.0 years ($SD=1.0$ years) and 68% were females. Participants gave informed consent and were compensated with either \$10 or partial course credit for their approximately 1 hr participation.

Materials

Prior knowledge. Knowledge tests were adopted from a previous study on vaccine misconceptions (Trevors & Kendeou, 2020). Participants were first asked to complete a prior knowledge pre-test to assess their misconceptions about vaccines. The test consisted of nine True/False items on vaccines with additional items on other socio-scientific issues so as to not bias participants with foreknowledge of the topic of the study. Each of the nine vaccine items was specific to a misconception that was addressed by one of the texts used in the study. One point was awarded for a correct answer and zero points for an incorrect answer.

Participants' prior knowledge of correct vaccine concepts was low: overall average prior knowledge score was 40% ($SD=27\%$) when corrected for guessing. Cronbach's alpha was also low ($\alpha=.50$), which taken together, demonstrated the presence of an erroneous and fragmented mental model of vaccines that may also be indicative of a flawed mental model (Gadgil, Nokes-Malach, & Chi, 2012). To further validate the prior knowledge test, we also assessed prior attitudes with an eight-item survey (e.g., "I approve of vaccines"; $\alpha=.77$), which correlated with prior knowledge, $r=.60$, $p<.001$.

Emotion induction. Participants were randomly assigned to the positive, negative, or no emotion induction (neutral) condition. Induction consisted of watching approximately 12 min of popular culture videos previously validated to induce either positive or negative emotional experiences (Bohn-Gettler & Rapp, 2011; Rottenberg, Ray, & Gross, 2007). Participants assigned to the positive induction condition watched video segments from "Whose Line is It Anyway?" and participants assigned to the negative induction condition watched segments from films such as "Bambi" and "The Lion King." There were no videos presented to those assigned to the neutral condition. In individual sessions, videos were presented on a computer screen while participants watched and listened with noise-cancelling headphones.

Positive and Negative Affect Scale. To assess treatment fidelity of the emotion induction procedure, participants responded to the Positive and Negative Affect Scale (PANAS-X; Watson & Clark, 1994) immediately before and after watching the videos of their respective experimental conditions. The PANAS-X is a questionnaire that assesses participants' current emotional states consisting of several subscales. Four subscales were relevant for the current study: positive and negative affect, sadness, and joviality. The questionnaire consists of a list of emotion words (e.g., positive affect: "inspired"; joviality: "delighted"; negative affect: "distressed"; sadness: "sad") and participants are asked to indicate to what extent they feel each emotion presently along a 5-point Likert-type scale ("Very slightly or not at all"–"Extremely"). In the current study, the four affective subscales had Cronbach's alpha coefficient that ranged between .80 and .93 (pre-induction) and .83 and .95 (post-induction).

Texts. Texts and design were adopted from a prior study (Trevors & Kendeou, 2020) and consisted of nine narrative-informational documents, each correcting one common misconception about vaccines (e.g., "multiple vaccines won't overwhelm the body") in a dialogue between two characters (see Supplemental Appendix A for an example text). All texts began with an introduction section that established the context of the narrative ($M=88$ words, $SD=9$). Following this, one of three within-subjects experimental conditions was presented. The Refutation + Explanation condition

consisted of stating and refuting the targeted incorrect belief ($M=38$ words, $SD=4$) and an explanation of the correct knowledge ($M=89$ words, $SD=6$). The Refutation-Only consisted of the same refutation section but without the explanation section. In lieu of an explanation, narrative content was presented that continued the story line without mentioning either correct or incorrect knowledge and was roughly equivalent in length (non-explanation section: $M=99$ words, $SD=9$). The No Refutation + No Explanation control condition presented narrative continuation equivalent in length to the refutation section (non-refutation section: $M=44$ words, $SD=10$) and the same non-explanation section as the Refutation-Only condition. In all conditions, a filler section followed that progressed the narrative without informational content ($M=57$ words, $SD=6$). After the filler sections, across all conditions, a correct-outcome sentence was presented that succinctly stated the correct knowledge. Reading times were recorded in milliseconds on the correct-outcome sentences. All correct-outcome sentences were within the range of 39–43 characters long ($M=41.7$). Finally, all texts ended with a conclusion to the story ($M=32$ words, $SD=5$). To ensure participants were reading for comprehension, a Yes/No comprehension question was presented that required a response before proceeding onto the next text. The comprehension questions focused on basic plot details and not correct or incorrect knowledge.

Three material sets were constructed wherein each set contained the nine texts with three texts in each of the three experimental conditions. Across the three material sets, each text was counter-balanced to occur once in one of the three conditions. Participants were randomly assigned to a set.

Post-test. After reading, participants were asked to complete the same nine-item True/False knowledge test from the start of the study. Following each True/False question, a second assessment tier was added that required participants to provide a brief, open-ended explanation for their answer. For the first True/False tier, one point was awarded for a correct answer and zero points for an incorrect answer. For the second explanation tier, two points were awarded for a complete and correct explanation, one point for a partially complete/correct explanation, and zero points for an incorrect explanation. It was possible for participants to answer the True/False question incorrectly yet still provide a correct explanation. Scores were summed to give a measure of post-reading knowledge. The possible range of scores was, therefore, 0–3 on each question and 0–9 within each text condition. Overall reliability was acceptable (Cronbach's $\alpha=.70$).

Procedure

Participants were greeted and informed that they would be watching videos and reading texts on a computer as part of a study on reading comprehension. All participants first

completed the prior knowledge pre-test and initial administration of the PANAS-X. Next, participants in the emotion induction conditions were told that they were going to watch a few video clips. They were requested to focus on the videos and allow themselves to become immersed and feel the emotions that the film scenes were conveying. Participants watched the videos in a private laboratory setting with noise-cancelling headphones. After watching, induction participants completed the PANAS-X for a second time and proceeded to reading the experimental texts. In lieu of videos, neutral condition participants proceeded directly to experimental texts.

Following Trevors and Kendeou (2020), each of the nine text trials began with the word READY in the centre of the screen and consisted of one whole text (Introduction to Conclusion) that was segmented into lines of approximately seven words each. Participants were instructed to read for comprehension at their normal pace. They were also instructed to rest their dominant hand on the space bar of the keyboard and to press it after they finished reading one line and were ready to move on to the next. Reading time was measured as the time between presses using E-Prime (v. 2.0). Once participants finished reading a text, the word QUESTIONS appeared in the centre of the screen for 2,000 ms followed by a comprehension check. Participants responded by pressing a key that corresponded with either “Yes” or “No.” Incorrect responses resulted in the word ERROR appearing in the middle of the screen for 750 ms. To ensure participants were familiar with the experimental procedure, participants went through three practice trials prior to the experimental refutations.

After reading all texts, participants completed the nine-item, two-tiered post-test and were instructed to answer to the best of their knowledge. Following the post-test, participants completed a short demographics questionnaire. Participants who received the negative emotion induction were shown the positive induction videos prior to their departure.

Results

Treatment fidelity

Induction condition (positive, negative, neutral) was entered as an independent variable in a multivariate analysis of variance (MANOVA) with PANAS-X subscale scores (positive affect, negative affect) measured at baseline (pre-induction) entered as dependent variables. Results showed that emotions did not differ between conditions at baseline, $F(8, 182)=1.70$, $p=.10$. However, after induction, emotion subscale scores significantly differed in the expected directions (Figure 1): positively induced participants reported higher positive affect, $t(53)=6.03$, $p<.001$, $d=1.62$, and joviality, $t(53)=10.17$, $p<.001$, $d=2.74$, and lower negative affect, $t(53)=-3.71$, $p<.001$,

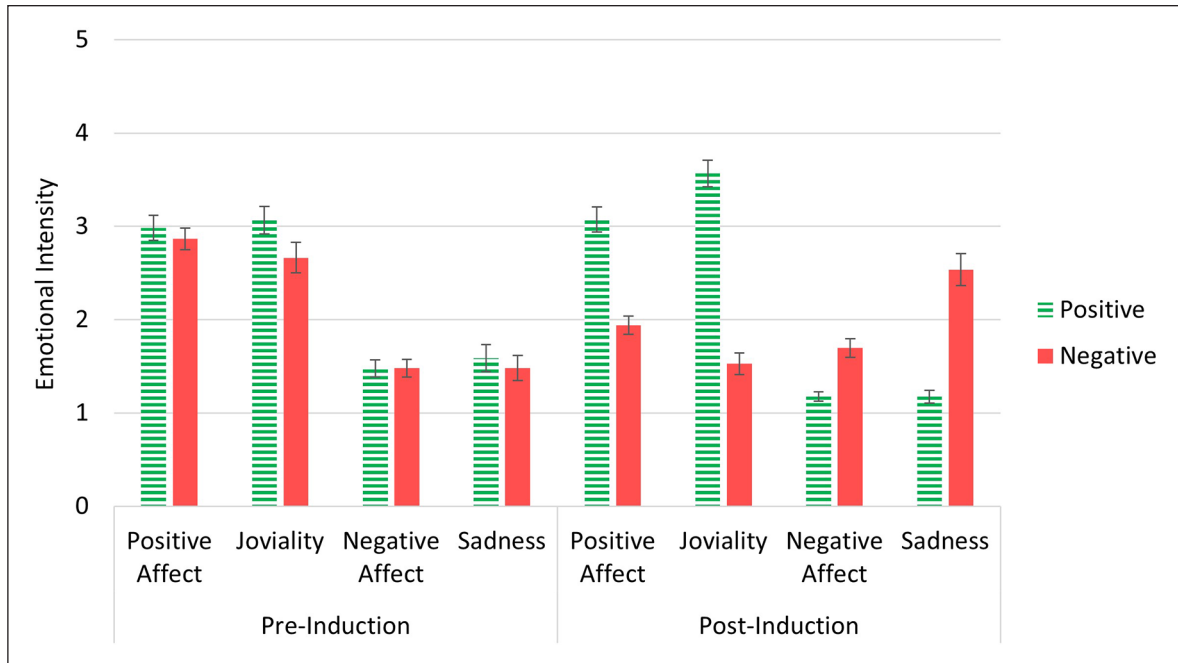


Figure 1. Self-reported intensity ratings (range, 0–5) of four emotional subscales from the PANAS-X measured immediately before and after emotion induction procedure. Experimental induction conditions were equivalent in their emotional states at pre-induction but differed in the intended direction at post-induction: positively induced participants reported significantly higher positive affect and joviality and lower negative affect and sadness compared to their negatively induced counterparts. Error bars denote 95% confidence intervals.

$d = 1.01$, and sadness, $t(53) = -6.47, p < .001, d = 1.76$, than their negatively induced counterparts. Large effect sizes demonstrate that the emotion induction procedure successfully produced the intended emotional states.

Reading times

Reading times of outcome sentences that were greater than 2.5 SDs from the mean or less than 750 ms were discarded. Furthermore, data for participants with reading times that exceeded these limits at four or more instances or failed four or more comprehension checks were discarded. As a result, outlier data from 11 participants were excluded.

Descriptive statistics for reading times are depicted in Figure 2. Text condition (Refutation + Explanation; Refutation-Only; No Refutation + No Explanation) and Induction condition were entered into a mixed effects ANOVA along with material set, entered as a blocking variable. The results showed a significant main effect of Text, $F(2, 152) = 10.77, p < .001, \eta^2 = .124$. Consistent with hypothesis related to text structure, correct outcome sentences were read significantly faster when presented in Refutation + Explanation than either Refutation-Only, $t(84) = 3.01, p = .003, d = .33$, or No Refutation + No Explanation texts, $t(84) = 3.30, p < .001, d = .36$, there was no difference between Refutation-Only and No Refutation + No Explanation texts, $p > .80$. Contrary to

our hypothesis related to emotion, there was no significant main effect for Induction condition, $F(2, 83) = 1.93, p = .15$.

Consistent with the hypothesis related to interaction effects, there was a significant interaction between Text $\times$ Induction conditions on reading times of correct outcome sentences, $F(4, 152) = 3.61, p = .008, \eta^2 = .087$. Specifically, in the Refutation-Only text condition, planned comparison showed that negatively induced participants read outcome sentences faster than positively induced participants, $t(53) = 3.14, p = .003, d = .85$.

Post-test scores

Descriptive statistics for post-test scores are depicted in Figure 3. A mixed effects ANOVA showed a significant main effect of Text condition, $F(2, 152) = 14.02, p < .001, \eta^2 = .156$. In support of our hypothesis related to text structure, planned comparisons showed that scores for items associated with Refutation + Explanation texts were higher than both Refutation-Only, $t(84) = 4.82, p < .001, d = .52$, or No Refutation + No Explanation, $t(84) = 3.95, p < .001, d = .43$. Furthermore, consistent with hypotheses related to emotion, there was an Induction condition main effect, $F(2, 76) = 3.47, p = .036, \eta^2 = .084$, such that negatively induced participants scored higher than positively induced participants overall, $t(54) = 2.62, p = .011, d = .71$. Contrary to our hypothesis related to interaction effects, there was no

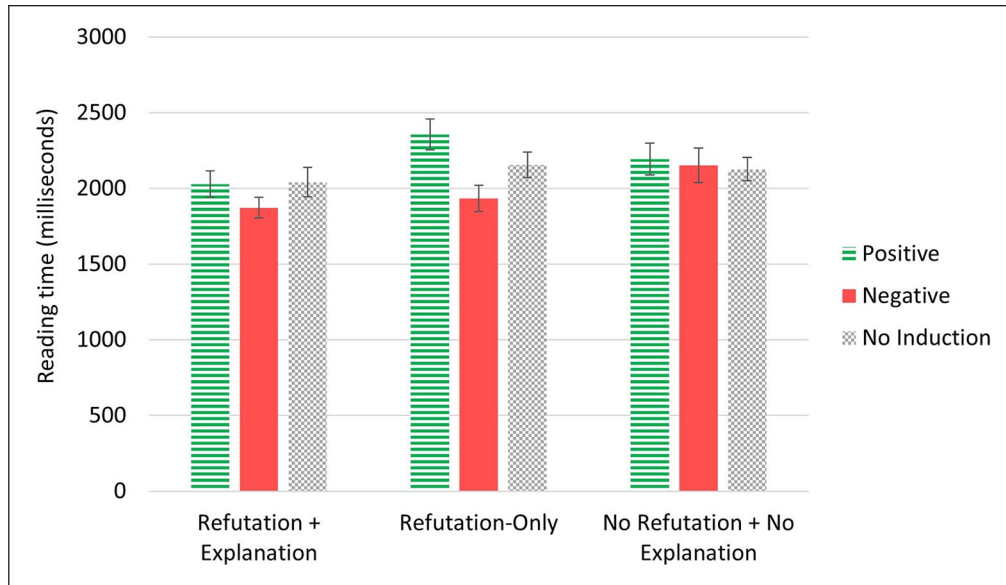


Figure 2. Reading times for correct outcome sentences in milliseconds within three text conditions and between three experimental emotion induction conditions. Error bars denote 95% confidence intervals.

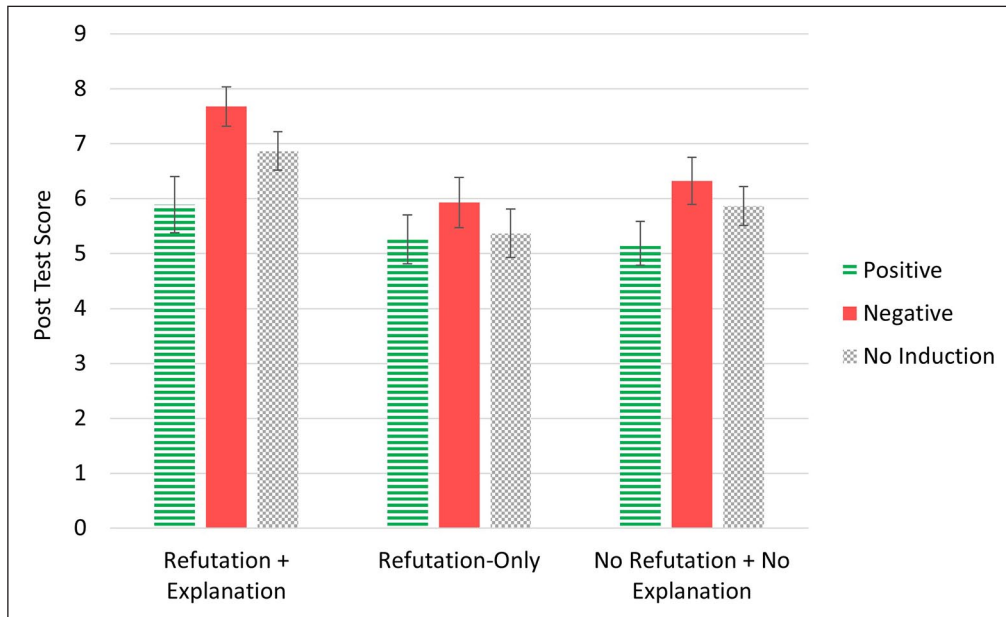


Figure 3. Knowledge post-test scores (range, 0–9) within three text conditions and between three experimental emotion induction conditions. Error bars denote 95% confidence intervals.

significant interaction between Text $\times$ Induction conditions, $F(4, 166) = .72, p = .58$.

Discussion

The current study is the first, to our knowledge, to experimentally examine the effects of readers'-induced emotions on the processes and outcomes of knowledge revision.

With respect to cognitive processing, text structure showed the expected effects, wherein refutations with explanations resulted in the fastest reading times of key text content overall. This reflects individuals' greater ease at integrating correct content into their mental representations after reading these texts, because the addition of the causal explanation presumably diminished the reactivation and interference of competing misconceptions. Emotion was

found to have an effect on processing of refutations without explanations, as the hybrid design of these texts may leave room for heterogeneous responses more susceptible to the influence of emotion. Specifically, negative emotion induction resulted in faster reading times of a critical part of the Refutation-Only texts (i.e., the correct outcome sentences), compared with positive emotion induction. With respect to learning, both text structure and emotions had independent main effects as expected, yet no similar interaction between text and emotion induction condition was observed. Refutations with explanations and negative emotions separately resulted in higher learning scores compared with other text structures and positive emotions, respectively. The differences in the pattern of effects between the two ways of measuring knowledge revision (processes and outcomes) warrants further investigation.

Even though the text structure effects were consistent on processing and outcomes of reading, confirming the hypothesis of the superiority of Refutation + Explanation texts, the influence of emotions appeared less consistent. Specifically, there was an interaction between induction and text on processing and a main effect of induction alone on outcomes, partially confirming the hypothesis for the facilitative effects of negative emotion. We believe this may be, in part, due to different ways of measuring knowledge revision (reading time paradigm and post-reading memory test). Negative emotions may have indeed increased attention and analytical processing of correct refutation and explanation content, whereas positive emotions may have led to less focused processing. This may have resulted in negative emotional states facilitating greater activation of correct content, which enabled it to be more readily integrated and grow dominant in the conceptual network that contained misconceptions (Kendeou et al., 2013, 2014, 2019). Our reading time paradigm detected a moderate difference between positive and negative emotional groups (that may reflect a difference in activation on correct content) with Refutation-Only texts, such a difference may be eliminated in the Refutation + Explanation condition because these texts present a preponderance of correct information equally for both groups. Induction did result in a main effect on post-test performance after reading has been completed, which may be attributable to improved memory of correct information for those in a negative state. Overall, the findings align with a broad literature that negative emotional states experienced in situ may enhance processes and outcomes relevant to knowledge revision (Bohn-Gettler, 2019; Fiedler & Beier, 2014; Forgas, 2013).

The facilitative effect of negative emotions also aligns with research on mood congruency effects. In general, mood congruency describes a phenomenon where information that has an emotional valence that matches the valence of individuals' current emotional state will be more readily attended to and encoded in memory (Becker & Leininger, 2011; Chang et al., 2019; Schuch & Koch, 2015; Yiend,

2010). Refutations create cognitive conflict and conflicting stimuli, in general, are perceived as possessing a negative emotional valence (Botvinick, 2007; Dreisbach & Fischer, 2012; van Steenbergen, Band, & Hommel, 2010). Thus, conflicting information may receive privileged processing and encoding in a matching negative state. This advantage may explain why negatively induced participants displayed faster reading time of correct outcome sentences in the Refutation-Only condition: a match between emotional stimuli and state resulted in refutation content being more activated in memory, facilitating the integration of correct outcome sentence. This explanation, however, rests on the assumption that refutation segments were indeed perceived as negatively valenced. We examined this assumption in a follow-up study.

Study 2

The goal of Study 2 was to assess whether refutation segments were perceived as more negatively valenced than equivalent content, which would support a congruency effect account for results obtained in Study 1.

Method

Participants. Participants were 35 US adults recruited via Amazon's mTurk platform ($N=35$). Their average age was 36.0 years ($SD=11.2$ years) and 60% were males. Participants gave informed consent and were compensated with \$2 for their approximately 17 min participation.

Materials

Texts. Participants were presented with sections of the same texts from Study 1; specifically, participants were presented with 18 sections that relayed 9 refutation sections and 9 analogous non-refutation sections.

Ratings. Presented below each text segment were three items that asked participants to rate segments according to how important/unimportant (perceived importance), emotionally negative/positive (emotion valence), and emotionally low energy/high energy (emotion activation) it appeared. Items were rated along a 7-point Likert-type scale.

Procedure. Participants who met qualification requirements (approval of past performance of 97% or higher, current location within the United States) were recruited. Responses were screened and those that failed to meet requirements were removed ($n=3$) and replacements recruited.

Results

Descriptive statistics for ratings are presented in Table 1. Three paired sample *t*-tests were conducted between refutation and non-refutation segments on perceived importance,

Table 1. Means and standard errors for refutation and non-refutation segment ratings.

	Importance	Valence	Activation
Refutation	5.65 (.15)	3.38 (.12)	4.44 (.17)
Non-refutation	4.68 (.17)	4.15 (.10)	3.29 (.18)

Possible range = 0–7; higher scores denote higher perceived importance, positivity, and activation of segment. Standard errors in parentheses.

emotion valence, and emotion activation. Results showed that refutation sentences were rated as more important, $t(34)=7.19$, $p<.001$, $d=1.22$, negative, $t(34)=-6.17$, $p<.001$, $d=1.04$, and activating, $t(34)=6.75$, $p<.001$, $d=1.14$, than their non-refutation counterparts.

Discussion

Findings from Study 2 provide preliminary support for a mood congruency effect. The emotional valence of refutation segments was perceived as more negative than equivalent non-refutation content. In addition, refutation segments were also perceived as more important and emotionally activating. All three dimensions likely contribute to a relative advantageous processing of correct content relayed by refutation texts. Furthermore, the congruency between perceived negative valence and negatively induced participants might account for the findings observed in Study 1. Nevertheless, the results from Study 2 do not rule out other potential synergistic effects and explanations. We address these in the general discussion.

General discussion

This study is among the first to investigate the effects of experimentally induced emotions on revising misconceptions. Revision was assessed with two methods, one focusing on cognitive processes online via reading times and one focusing on learning outcomes after reading offline via post-test of knowledge. Across both methods, a negative emotional state improved knowledge revision compared with a positive state. These results may be, in part, driven by accommodative processes and/or congruency effects between individuals' negative emotional state and the negatively valenced refutation content, and they can be interpreted in light of contemporary frameworks of knowledge revision (Kendeou et al., 2013, 2014) and the influence of emotions on comprehension (Bohn-Gettler, 2019).

Refutation texts and knowledge revision

The current findings replicate the effects of refutation texts overall (Ariasi et al., 2017; Diakidoy et al., 2016; Kendeou et al., 2013, 2014, 2019; Thacker et al., 2020; Trevors et al., 2017; van den Broek, 2010; van Loon et al., 2015).

Refutation sentences paired with extended explanations (Refutation + Explanation) resulted in faster reading times of correct outcome sentences and higher post-test scores compared with refutations without explanations (Refutation-Only) or control texts (No Refutation + No Explanation). Reading times of correct outcomes are indicative of the ease of integrating the correct concept into readers' mental representations (Kaakinen, 2017; Kendeou et al., 2013, 2019; Murray & Engle, 2005). Reading times on select content provide a window into individuals' current mental models. Slower times indicate that prior misconceptions continue to be active, conflict with, and disrupt integration, whereas faster times indicate that misconceptions are no longer activated or interfering with comprehension of new accurate content. This is due to the new addition of correct explanations to readers' mental representations introduced by refutations with explanations that competitively draw a greater share of limited activation at the expense of prior misconceptions. Higher post-test scores indicate that individuals are able to recall the correct information when choosing and constructing post-test responses. In sum, texts with refutations and explanations of vaccine misconceptions replicated both process and outcome effects obtained in the extant literature. This validates the effectiveness of the current materials and permits conclusions to be drawn about the differences observed when emotions were introduced and manipulated.

Influence of emotions. Compared with positive emotions, being in a negative emotional state facilitated knowledge revision processes (under certain conditions) and accuracy after reading. Notably, during reading, this effect was only observed in the Refutation-Only condition, whereas after reading it was evinced in all text conditions. There are at least two explanations for these effects.

One explanation rests on the assumption that both the Refutation-Only condition and the open-ended post-test format are types of tasks that are much less constrained than either of the other two text conditions or a hypothetical closed-ended test format, respectively, and thus more susceptible to the influence of emotion (Bohn-Gettler, 2019). Indeed, the Refutation + Explanation condition supports a specific sequence of cognitive processes that restricts the activation of misconceptions and instead privileges allocation of activation towards correct concepts (Kendeou et al., 2013, 2014, 2019), leaving less room for other factors like emotions to influence activation. In contrast, the Refutation-Only texts co-activate correct and incorrect concepts, letting them both linger, but does not resolve the competition with additional explanations to direct activation towards either one. With respect to the current post-test, its scores reflect achievement on a two-tiered test consisting of true/false questions and writing open-ended explanations, which also reflects a less

constrained task. Together, with fewer external constraints provided by tasks (texts or tests), emotion may be an influential factor that directs activation, which we hypothesised to differ depending on valence. According to the PET framework, under these circumstances, negative emotions facilitate bottom-up accommodative processing to change pre-existing concepts, such as prior misconceptions, to reflect new information obtained from the external environment, such as a refutation text (Bohn-Gettler, 2019). In contrast, positive emotions promote top-down assimilative processes to reinterpret new information to fit into pre-existing conceptual structures. Thus, negatively induced participants may have integrated refutations to a greater extent as their positively induced counterparts, and this integration had an added benefit during reading when the task was unconstrained. This explanation is also partially consistent with the effects obtained after reading: while an interaction between condition and emotion was not observed, across all three text conditions, negatively induced participants attained the highest scores on an open-ended post-test that was similarly less constrained. Together, negative emotion induction paired with less constrained tasks may have resulted in greater accommodative processes that are integral to revising misconceptions.

A second explanation rests on the assumption of congruency. As the findings from Study 2 showed, refutation content was perceived as possessing a negative valence. Negatively valenced information generally captures greater levels of attention and promotes better memory than positive or neutral stimuli (Baumeister, Bratslavsky, Finkenauer, & Vohs, 2001; Chang et al., 2019; Estes & Adelman, 2008; Mather & Sutherland, 2011). In the current study, refutations may, therefore, be more salient in general, which may be further amplified for individuals in a congruent (negative) versus incongruent (positive) emotional state, reflecting a congruency effect (Becker & Leininger, 2011). Heightened salience of correct refutation content at its initial encoding may have resulted in a higher level of activation that eased integration of parallel correct outcome sentences later on in the text (Kendeou et al., 2019) and improved memory of refutation content reflected by higher post-test scores after reading. Therefore, the current findings are consistent with past research suggesting that negative emotions are linked with increased salience, focused attention, and detailed-oriented processing, whereas positive emotions are linked with a broadened scope of attention and heightened memory for peripheral and potentially irrelevant information (Baas, De Dreu, & Nijstad, 2008; Bodenhausen, Gabriel, & Lineberger, 2000; Chang et al., 2019; Fredrickson, 2013; Gasper & Clore, 2002; Kensinger, 2009; Paul, Walentowska, Bakic, Dondaine, & Pourtois, 2017; Rowe, Hirsh, & Anderson, 2007; Schmidt, Rosga, Schatto, Breidenstein, & Schwabe, 2014; Senderecka, 2018; Storbeck & Clore, 2005; Vanlessen, Rossi, De Raedt, & Pourtois, 2014; Whitmer & Gotlib, 2013). However, the different pattern of effects of emotion between the two

methods of measuring knowledge revision (an interaction on revision processes and a main effect on revision outcomes) warrants further investigation.

Overall, the findings taken together provide new insights into the influence of emotion experienced in the moment on comprehension during revision of misconceptions. Corrections of misconceptions were perceived as possessing a negative valence and were differentially processed depending on the valence of individuals' own emotional state and structure of the refutation text. Process and product data converged to show that a positive state diminished, whereas a negative state amplified correct information in memory. The current findings may be the result of an accommodative and/or congruency effects. Although these explanations are not mutually exclusive, it would be important to identify which of these effects is the most dominant in future research. In sum, these results extend theory about knowledge revision by showing that individuals in the moment of emotional state will, in part, modulate the competition for limited activation between conflicting ideas, which determines whether misconceptions will persist, or correct understanding will dominate.

Limitations and future directions

The contribution of the current investigation is limited in several ways that suggests potential paths future research. First, although we hypothesise that emotions affect co-activation, integration and competition among conflicting concepts that is in part revealed with reading times as a behavioural measure, other observational methods of measuring these cognitive processes are advisable to acquire converging evidence. Second, consistent with a long tradition in affective science (Coan & Allen, 2007), we experimentally induced emotion to study its causal effects. However, such manipulation creates an emotional state that may be incidental to the content of the task, in contrast to emotions that naturally occur in response to the content, which may be more integral and interact differently with cognitive processes. Future research can build on the current study via examination of such endogenous emotions. Third, the influence of emotions on comprehension will vary as a function of tasks and processes (Bohn-Gettler, 2019) and specific type of discrete emotion (Weeks, 2015). Although the current effects were examined across several texts and misconceptions on the topic of vaccination, which itself may be perceived to carry its own unique valence, it will also be important to assess the generalisability of the current conclusions with other materials and topics as well.

Implications for practice

In sum, emotion influenced comprehension over the course of a text-based intervention that attempted to correct common vaccine misconceptions, wherein negative emotion enhanced, and positive emotion diminished cognitive

processes and outcomes of knowledge revision. The findings are practically relevant for educators and health professionals because they demonstrate that negative emotions are not invariably linked to failure. Rather, in situations that call for the revision of misconceptions in light of new, corrective evidence, appropriately regulating negative emotions during learning may confer an advantage insofar as they facilitate accommodative processes. Moreover, for some important topics that are addressed in refutation texts, such as vaccination, educators and health professionals may begin with a reminder of the potential risks of serious harm to individuals and society if misconceptions are left unaddressed, which itself may be a minor form of emotion induction. Finally, insofar as current findings suggest potential explanatory mechanisms (e.g., increased attention and integration of central refutation content), educators and health professionals may also scaffold and support such cognitive processes to promote knowledge revision as well. Overall, as the world prepares for a global vaccine-promotion campaign to gain control over the COVID-19 pandemic, the current findings contribute to practical and theoretical knowledge of correcting socially relevant misconceptions.

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Supplementary material

The supplementary material is available at qjep.sagepub.com.

Note

1. A statistical power analysis was performed to estimate sample size, based on a similar study measuring the effects of emotion induction on reading comprehension processes and outcomes from Bohn-Gettler and Rapp (2011). The median main effect size of emotion induction in this study was $\eta^2 = .11$ (range of $\eta^2 = .01-.81$). With $\alpha = .05$ and power = .80, projected sample size needed with the median effect size is approximately $N = 84$ (GPower 3.1.9.2).

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